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Utilization of traditional medicines in livestock health management in Manud block, Ramgarh, Jharkhand, India

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Abstract : A study was carried out in Mandu Block of Ramgarh District of Jharkhand state to document the role of traditional medicines in livestock health management amongst indigenous peasants and pastoralists. A cross sectional survey involved administration of questionnaire to the key informants and resource persons of all 31 villages of the Block.

The results indicated that 85% of the respondents regularly use traditional medicine in treatment of their livestock. There were 6 basic categories of diseases frequently treated. Dysentery, Foot and Mouth Disease and Boils & Sores were the most frequently treated diseases. Various ingredients were used along with herbs like Gur, Flour, Milk and Sodium bicarbonate etc.

The local Vaidya and elderly people of the village had rich knowledge regarding composition and administration of traditional drugs. It had been concluded that the indigenous people of Mandu Block were heavily dependent on herbal drugs for the treatment of variety of livestock ailments. Knowledge of traditional medicines always provides a base line for phytochemical and pharmacological investigation. The documentation of the knowledge regarding traditional medicine in study area has never been done before. So it is necessary to record these knowledge before it is too late.

Keywords : Traditional medicine, Livestock, Mandu Block, Resource persons, Key informants, Pharmacological.

INTRODUCTION

The role of traditional medicines in livestock health management is beyond dispute[1]. It has been existed alongside human evolutionary history, taking many different forms. A great number of professionals from varied fields have over the past several years recognised, valued, documented and ethnocentrically studied the potential effectiveness of the livestock health care practices embodied in native and local communities. The studies focused on a holistic evaluation of the traditional and non-traditional knowledge, which hold some beneficial aspects relevant to the improvement and development of world poorest rural communities. Comprising all ethnopractical approaches and traditional knowledge applied by human with a view to alleviating health complaints afflicting their livestock and hence, improves their production and performance. A glance at the existing literature reveals that the traditional knowledge embodied in ethnomedicine, constitute yet an untapped resources of potentially useful information for possible deployment in sustainable animal health management systems in rural and peri-urban communities all over the world[2,3]. The study may take the form of selective breeds and breeding practices, crowning and recognition of renowned ethnopractitioners, animal feeds and feeding behaviour, ritualism, herbalism, spiritualism, ethnoepidemiological knowledge on livestock vectors, pathogens, hosts and diseases and traditional "institutions" and ecosystems in which this knowledge exist.

Since pre-historic times, man was aware of traditional veterinary medicine, though haphazardly studied and recorded, until recently. A milestone in understanding the traditional veterinary medicine as applied and used by local people, was the determination and definition of the term ethnoveterinary. This term was for the first time in research forum, applied and used by Dr. Constance McCorkle in

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1986[4]. She used it to refer to the systematic investigations of folk beliefs and practices in veterinary medicine; by then and later, defined it as local or indigenous knowledge and methods for caring for, healing and managing livestock. This included social practices and ways in which livestock are incorporated in farming system. Like any other knowledge system ethnoveterinary knowledge is very dynamic in its evolution, management and practice. Because of this dynamism, many ethnopractitioners find themselves in a situation where they complement ethnoveterinary knowledge with modern veterinary medicine, especially in case where ethnoveterinary knowledge is limited and or cannot work[5].

Livestock have revolutionised the rural economy of India[6]. It plays an integral and significant role in smallholder subsistence in diverse states of India. In Jharkhand the majority of farming households keep a range of livestock including cattle, buffaloes, goats and pigs which form a traditional role in their livelihoods. Livestock sector have the capacity to provide opportunities for livelihood to people at the place where and in the situation they are. Growing demand for livestock and its products in the urban and rural areas of India emphasizes the opportunity for increased livestock development initiatives. Livestock production has the potential to become an economic enterprise that targets the poor and marginalised peasants[7].

STUDY AREA

Mandu block is situated in Ramgrah District, and lies between latitude 23^0 38' 553 to 23^0 58' 403 North and longitude 85^0 25'443 to 85^0 39' 373 East, having 6,445.40 acre total area which comprises of 45,086.85 acre of forest land and 1,858.55 acre of bare land. Among the scheduled castes mainly Ganju, Rajwar, Ravidas and Rajaks are prevalent. The tribal population of the Mandu block comprises of Santhals, Bedia, Karmali, Munda, Oraon and Birhore. Common people generally depend upon the traditional medicines for treatment of their livestock in Mandu Block.Use of various plant parts and their knowledge of traditional medicines in livestock health management is invaluable which needs to be documented.

MATERIALS AND METHOD

Field work was conducted from May 2014 to November 2015 with the help of B.D.O. and Forest Officer of Mandu Block. Frequent visits were paid to Vaidyas, Pahans and Horopaths having strong traditional knowledge regarding livestock treatment. Semistructured questionnaires were designed addressing detailed ethnoveterinary information in which they were asked about the name of plants they use to treat their livestock, the part of plants used, recipe formulation and its recovery. The voucher specimens collected have been deposited in the Ethnobotanical laboratory, P.G. Department of Botany, Ranchi University, Ranchi after proper scientific identification.

OBSERVATIONS

33 plants have been enumerated in alphabetical order with their scientific name, family and local name. The herbal remedies are in their actual form what the respondents are regularly using to cure livestock from variety of ailments.

1. *Abrusprecatorius* L. (Fabaceae) "Ratti", "Gunci" RKRM – 006
   Paste of 6 to 8 gram seeds is fed orally to cattle for expulsion of placenta. [Bedia]

2. *Acacianilotica*(L.) Delile (Mimosaceae) "Babul" RKRM – 012
   Dried bark of plant is boiled in water and filtered through fine cotton cloth. Filtrate is given to cattle thrice a day to cure gastric problems. [Santhal]

3. *Acoruscalamus*L. (Acoraceae) "Bach" RKRM - 018
   Cattles are relieved from the menace of lice and flies by periodic use of powdered rhizomes. [Karmali, Santhal]

4. *Alliumsativum*L. (Alliaceae) RKRM – 026
   2 to 3 bulblets are given to ships with its feed to get rid of parasitic worms. [Ganju]

5. *Aloevera*(L.) Burm.f. (Aloaceae) "Ghikumari" RKRM – 030
   Herbal drug containing turmeric, aloevera, gur, common salt, garlic and coconut extract is given to cure Foot and Mouth Disease. A dose of 50 ml is administered orally to the affected cattle and 30 ml to the sheep and calves. [Oraon]

6. *Annonasquamosa*Delile (Annonaceae) "Sharifa" RKRM – 032

Powdered seeds and crushed immature fruits are used against fly larvae and other insects. [Birhore]

7. *Azadirachtaindica*A.Juss. (Meliaceae) "Neem" RKRM – 036
Burnt leaves mixed with coconut oil speeds up the healing of wounds.

8. *Balanitesroxburghii*Planch. (Balanitaceae) "Hingota" RKRM – 039
Fruit paste aids in expulsion of placenta during parturition in cattle folks. [Munda, Bedia]

9. *Brassicacampestris*L. (Brassicaceae) "Sarson" RKRM – 049
12 gm. Asafoetida and 50 gm. Garlic paste is mixed with half kg mustard oil. Mixture is given twice a day cattles to get rid of gastric problems. [Karmali, Bedia]

10. *Buteamonosperma*(Lam.) Taub. (Fabaceae) "Palas" RKRM – 081
Grinded seeds are given to sheep with its feed to get rid of mixed infections of gastrointestinal roundworms. [Santhal, Birhore]

11. *Calotropisprocera*W.T.Aiton (Asclepiadaceae) "Akwan" RKRM – 093
Traditional balm is prepared by mixing stem ash and ghee. It is applied on wounds for early healing. 50 to 60 flowers are boiled in 1 litre of water and filtrate is given to goats once a day for 4 to 5 days to get rid of gastro-intestinal parasitic worm. [Santhal, Oraon]

12. *Cannabissativa*L. (Cannabaceae) "Bhang" RKRM – 111
Crude leaves extract is administered against flukes. [Ganju, Karmali]

13. *Capparisdecidua*Edgew. (Capparaceae) "Keir" RKRM – 116
Root charcoal mixed with 'Til' oil is given to cattle orally for speedy healing of fractured bones. [Munda,Santhal]

14. *Chenopodiumalbum*L. (Chenopodiaceae) "Bettusag" RKRM – 119
Decoction of whole plant is administered against gastro-intestinal worms in sheep. [Munda]

15. *Curcumalonga*L. (Zinziberaceae) "Haldi" RKRM – 124
Powdered rhizome is applied topically to cure boils and sores in cattle.[Munda, Bedia]

16. *Dendrocalamusstrictus*Nees (Poaceae) "Bans" RKRM – 138
In case of fractured bone of legs of cattle, strips of bamboo are tied over it for quick recovery. Water is sprinkled frequently on it to keep it moistened. [Santhal]
Decoction of bamboo stem is administered for expulsion of placenta in cattle. [Bedia]

17. *Dioscorea bulbifera*L. (Dioscoreaceae) "Barha kanda" RKRM – 141
Dried and pounded tubers are prescribed twice a day in case of dysentery and diarrhoea. [Karmali]

18. *Euphorbiahirta*L. (Euphorbiaceae) "Bara dudhi" RKRM – 146
Decoction of whole plant is given thrice a day orally to cattle to cure dysentery. [Oraon]

19. *Evolvulusalsinoides*L. (Convolvulaceae) "Samkranta" RKRM – 149
A balm is prepared by crushing lower part of plant including roots 5 part; black peepers 1 part; mustard oil 3 part; then applied to cure the sores on the neck of cattle caused during ploughing. [Karmali, Bedia]

20. *Helicteresisora*L. (Sterculiaceae) "Maror Phalli" RKRM – 158
100 gm fried pods are given to calves twice a day to kill intestinal worm.
50 ml decoction of bark is prescribed thrice a day to cure diarrhoea. [Oraon]

21. *Holarrhenapubescens*Wall. & G.Don (Apocynaceae) "Kurci" RKRM – 164
The wounds are cleansed with the decoction of its bark to promote healing in cattle. [Karmali]

22. *Meliaazedarach*L. (Meliaceae) "Bakayan" RKRM – 175
Extract of fruit is given to cattle to get rid of tapeworm. [Munda, Bedia]

23. *Menthapiperita*L. (Lamiaceae) "Peppermint" RKRM – 178
Essential oil concentrate is applied externally to save cattle from external parasites. [Munda]

24. *Pedaliummurex*L. (Pedaliaceae) " Bara gokhru" RKRM – 189
Gastro-intestinal problems are cured by feeding whole plant to cattle.[Santhal, Bedia]

25. *Pennisetum typhoidis* Burm. F. (Poaceae) "Bajra" RKRM – 196
Skin itching in cattle is cured by applying aqueous paste of inflorescence of this plant. [Santhal]

26. *Pongamiapinnata*(L.) Pierre (Fabaceae) "Karanj" RKRM - 201
Fruits oil applied direct on wounds of cattle for fast recovery without decaying. [Santhal]

27. *Prosopisspicigera*L. (Leguminosae) "Shami" RKRM – 214
During rainy season, affected foot of cattle is cured by applying aqueous paste of leaves. [Sathal]

28. *Ricinuscommunis*L. (Euphorbiaceae) "Reri" RKRM – 221
One teaspoon of seed oil is fed to cattle to cure stomach disorder and to sheep to control throat problems. [Santhal]

29. *Sesamumindicum*L. (Pedaliaceae) "Til" RKRM – 228
Turmeric powder mixed with hot seed oil is applied on foot to cure infection.
250 gm of seed oil is administered orally to cure dysentery. [Santhal]

30. *Semecarpusanacardium*L.f. (Anacardiaceae) "Bhela" RKRM – 234
Oil is extracted by heating cashewnut (*Anacardium occidentale*) shell (about 500g), 12 to 15 seeds of *Semecarpus anacardium* and 10g fresh green tender bark of teakwood (*Tectona grandis*). This oil is used to treat the parts of the animals infected by houseflies. [Bedia]

31. *Tinosporacordifolia*(Willd.) Miers ex Hook.f. & Thomson (Menispermaceae) "Giloy" RKRM – 238
Leaf paste with turmeric powder and pinch of salt is applied externally on boils twice a day. [Bedia]
Infusion of the stem is put in chicken's drinking water in case of poultry disease. [Ganju]

32. *Triticumaestivum*L. (Poaceae) "Genhu" RKRM – 238
Roasted seed paste is mixed with fodder of goat to cure their gastric problems. [Bedia]

33. *Zinziber officinale* Rosc. (Zinziberaceae) "Adrak" RKRM – 251
Dried rhizome, rock salt, gur and sodium bicarbonate are mixed properly and mixture is given to cattle for 3-4 days to cure indigestion. [Munda]

RESULTS AND DISCUSSION

The present paper deals with 37 different prescriptions of ethno-veterinary usages of 33 plant species of 33 genera belonging to 25 families. Out of these, 13 plant species are for curing various types of gastrointestinal problems, followed by 5 plant species for getting rid of variety of external parasites and 3 for the healing of Boils & sores. Apart from these, there are other ailments of livestock like Foot and Mouth Disease, infestation of variety of worms, decaying of horns, which are treated very effectively by the local Vaidyas and knowledgeable persons. The study reveals that these easily available, cost effective, low priced traditional medicines are the only way for the treatment of their livestock because majority population are economically weak and availing modern high priced allopathic medicine will put extra pressure on their economy. It is also recorded that all these preparations are highly effective and without any side effect. Further research on herbal prescription may help in developing effective and curative drugs for proper health management of livestock.

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